

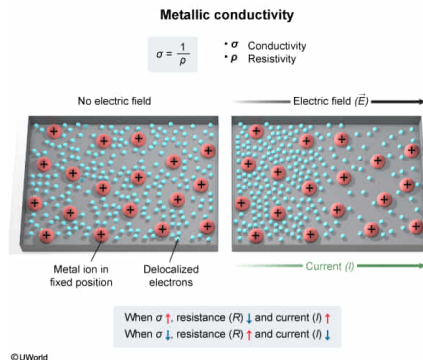
Suppose that a simple circuit comprising one voltage source and one metallic resistor yields current I . If the resistor were replaced with another resistor that is identical except that it has 75% lower conductivity, then I would:

- ☐ A. increase by a factor of 2. [9%]
- ☐ B. increase by a factor of 4. [30%]
- ☐ C. decrease by a factor of 2. [18%]
- ☒ D. decrease by a factor of 4. [40%]

Correct

41% answered correctly

Explanation:



Conductivity (σ) indicates the ease with which electrons flow within the molecular structure of a material. For example, metals conduct electricity more readily than nonmetals because many of the electrons contained within metals are very loosely associated with the positively charged nuclei. In contrast, the conductivity of **electrolytic solutions** is proportional to the molar concentration of charged atoms.

Conductivity is **inversely proportional** to **resistivity (ρ)**, a variable that quantifies the relative difficulty with which electrons flow through a material:

$$\sigma \propto \frac{1}{\rho}$$

Both resistivity and conductivity are intensive properties related to the behavior of materials rather than objects. Therefore, modifying the physical dimensions of an **electric conductor** may influence the rate at which **current** flows through the conductor, but not the underlying conductivity of the material from which the conductor is formed. Consequently, conductivity is directly proportional to the conductance of an electric conductor just as resistivity is directly proportional to resistance (R) of a resistor.

In this question, a voltage source generates a current (I) through one resistor in a simple circuit. Replacing this resistor with one that has 75% lower conductivity is equivalent to changing the conductivity to $\frac{1}{4}$ (ie, 25%) of the original value:

$$\sigma_{\text{new}} = \frac{\sigma_{\text{old}}}{4}$$

Because resistivity and conductivity are inversely proportional, the resistivity of the new resistor is increased by a factor of 4 compared to the original:

$$\rho_{\text{new}} = 4 \rho_{\text{old}}$$

Given that the resistor is otherwise identical (ie, the same physical dimensions), the resistance of the resistor will also increase by a factor of 4.

$$R_{\text{new}} = 4R_{\text{old}}$$

Ohm's law relates the current generated by such a simple circuit to the voltage (V) and resistance:

$$I = \frac{V}{R}$$

Hence:

$$I_{\text{new}} = \frac{I_{\text{old}}}{4}$$

Therefore, replacing the original resistor with another that has $\frac{1}{4}$ the conductivity (4 times the resistance and resistivity) of the original will *decrease the current* traveling through the circuit to $\frac{1}{4}$ its original value (**Choice D**).

Educational objective:

Conductivity is an intensive property that quantifies the ease with which current flows through a material acting as an electric conductor. Conductivity is inversely proportional to resistivity.

Time spent:0 sec

Subject:
Physics

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System:
Electrostatics and Circuits